Levoxyl® comes in different dosages with different additives listed below.	
Strength (µg)	Color additive(s)
200	D&C Red No. 30 aluminum lake, D&C Yellow No. 10 aluminum lake

Unithroid®

Unithroid®: levothyroxine sodium tablets. It contains synthetic crystalline L-3,3',5,5'-tetraiodothyronine sodium salt. Its inactive ingredients are colloidal silicon dioxide, lactose, magnesium stearate, microcrystalline cellulose, corn starch, acacia, sodium starch glycolate, and dyes, based on tablet strength.

Note: Corn starch may be a cause of sensitivity for some, especially Hashimoto's patients who are gluten sensitive or have celiac disease. Acacia is a TH-2 stimulator and may cause immune responses with TH-2 dominant Hashimoto's patients. Lactose is a common sensitivity for many people.

Unithroid® comes in different dosages with different additives listed below.	
Strength (µg)	Color Additive(s)
25	FD&C Yellow No. 6 aluminum lake
50	None
75	FD&C Red No. 40 aluminum lake, FD&C Blue No. 2 aluminum lake
88	D&C Yellow No. 10 aluminum lake, FD&C Yellow No. 6 aluminum lake, FD&C Blue No. 1 aluminum lake
100	D&C Yellow No. 10 aluminum lake, FD&C Yellow No. 6 aluminum lake
112	D&C Red No. 27 aluminum lake
125	FD&C Yellow No. 6 aluminum lake, FD&C Red No. 40 aluminum lake, FD&C Blue No. 1 aluminum lake
150	FD&C Blue No. 2 aluminum lake
175	FD&C Blue No. 1 aluminum lake, D&C Red No. 27 aluminum lake
200	FD&C Red No. 40 aluminum lake
300	D&C Yellow No. 10 aluminum lake, FD&C Yellow No. 6 aluminum lake, FD&C Blue No. 1 aluminum lake

Cytomel®

Cytomel®: a liothyronine sodium tablet. It contains isolated synthetic T3. Its inactive ingredients are calcium sulfate, gelatin, starch, stearic acid, sucrose, and talc.

Cytomel® tablets: 5 µg in bottles of 100; 25 µg in bottles of 100; and 50 µg in bottles of 100. They are all white tablets without dyes. The dosage is clarified by imprints on the tables that list:

- 5 µg tablet imprinted with SKF/D14
- 25 µg tablet imprinted with SKF/D16
- 50 µg tablet imprinted with SKF/D17

Note: Inactive ingredients, such as starch, may be a hidden source of gluten.

Thyrolar®

Thyrolar®: liotrix tablets. It contains triiodothyronine (T3 liothyronine) sodium and tetraiodothyronine (T4 levothyroxine) sodium. The inactive ingredients are calcium phosphate, colloidal silicon dioxide, corn starch, lactose, and magnesium stearate. The tablets also contain the following dyes:

Thyrolar® comes in different dosages with different additives listed below.	
Strength (µg)	Color Additive(s)
¼	FD&C Blue No. 1 and FD&C Red No. 40
½	FD&C Red No. 40 and D&C Yellow No. 10
1	FD&C Red No. 40
2	FD&C Blue No. 1, FD&C Red No. 40, and D&C Yellow No. 10
3	FD&C Red No. 40 and D&C Yellow No. 10

Note: Corn starch may be a cause of sensitivity for some, especially Hashimoto's patients who are gluten sensitive or have celiac disease. Lactose is a common sensitivity for many people.

Armour®

Armour®: a natural preparation derived from porcine thyroid glands and has a strong, characteristic odor. It provides 38 µg T4 and 9 µg T3 per grain of thyroid. The inactive ingredients are calcium stearate, dextrose, microcrystalline cellulose, sodium starch glycolate, and Opadry® white.

Armour® tablets are evenly colored, light tan, round tablets, with convex surfaces. One side is debossed

Please see registered trademark ownership in Appendix 1.

with a mortar and pestle beneath the letter "A" on the top and strength code letters on the bottom as defined below.

Strength	Code
¼ grain	TC
½ grain	TD
1 grain	TE
1 ½ grain	TJ
2 grain	TF
3 grain	TG (bisected)
4 grain	TH
5 grain	TI (bisected)

Westhroid™

Westhroid™: "a natural preparation derived from porcine thyroid glands and has a strong, characteristic odor." It provides 38 µg T4 and 9 µg T3 per grain of thyroid. The inactive ingredients are microcrystalline cellulose, dicalcium phosphate, colloidal silicon dioxide, sodium starch glycolate, magnesium stearate, hydroxypropyl methylcellulose, stearic acid, carnauba wax, polyethylene, and lactose monohydrate.

Tablet Strengths
½ grain
1 grain
2 grain

Nature-Throid®

Nature-Throid®: a natural preparation derived from porcine thyroid glands and has a strong, characteristic odor. It provides T4 and T3. The inactive ingredients are microcrystalline cellulose, dicalcium phosphate, colloidal silicon dioxide, sodium starch glycolate, magnesium stearate, hydroxypropyl methylcellulose, stearic acid, carnauba wax, polyethylene, and lactose monohydrate.

Tablet Strengths
½ grain
1 grain
2 grain

Thyroid Hormones and Laboratory Testing

It is very important to understand the influences of synthetic versus bioidentical thyroid hormone replacement on thyroid hormone measurements. Simply put, laboratories do not measure synthetic hormones. Laboratories only measure bioidentical hormones. Therefore, if a patient is taking synthetic T4 or T3, they are not actually taking T4 or T3. They are taking a substance that is similar to the bioidentical structure, but different. For example, Synthroid® is levothyroxine sodium L-3,3',5,5'-tetraiodothyronine, and the empirical formula is C15H10I4N NaO4 • H2O. Actual T4 is 3,5,3',5'-tetraiodothyronine, and the molecular formula is C15H11I4NO4. They are not the same structure biochemically. They are similar enough to have a similar impact on metabolism, but not similar enough to account for laboratory measurements. Therefore, when patients take synthetic hormones, the hormones they are taking cannot be accurately measured. That is why when patients are placed on Synthroid® they are not evaluated by serum T4 testing, instead they are evaluated by TSH. That is because T4 testing does not measure what they are taking. The same could be said about synthetic T3. The bottom line is if a patient is taking bioidentical T3 and T4, the laboratory measurements of those hormones will be accounted for and measured in the test. However, if they are taking synthetic T3 or T4, the laboratory measurements of those hormones would not be accounted for and the test to use in these instances is TSH only.

Choosing Appropriate Thyroid Hormones

A question that always comes up is, "What is the best thyroid hormone?" The answer is, "It depends!" It depends on what the goal of therapy is. If the goal of therapy is to solely bring TSH within the laboratory range, any thyroid replacement can accomplish that. If the goal of replacement is to compensate for lack of T4-to-T3 conversion, then a T3-based replacement can be used. If the goal is to provide a bioidentical source of hormones, those are clearly identified. In reality, many thyroid patients, when working with a progressive physician (not a Synthroid®-exclusive practitioner), will find certain medications, brands, and combinations either work better, are worse, or cause little difference for them. So the answer is really experimentation in conjunction with subjective and objective monitoring.

Please see registered trademark ownership in Appendix 1.

There are some common issues that present themselves in clinical practice that need to be discussed. One of them is sensitivity reactions to certain medications. Some patients (especially autoimmune thyroid cases) may be sensitive to the dyes or fillers used in the tablets. A common mechanism of sensitivity may be the cornstarch found in fillers of many thyroid hormones. This is especially evident with Hashimoto's patients who are also gliadin-intolerant or have concurrent celiac disease. Corn appears to be similar to gliadin and can cause an immune cross-reaction process. Additionally, there is a small amount of gliadin in the protein structure of corn. For example, Synthroid®, Westhroid™, and Thyrolar® contain cornstarch. Armour® and Nature-Throid® do not. Overall, when all fillers and dyes are considered, Nature-Throid® appears to be the most hypoallergenic thyroid hormone. The chance that many patients, especially autoimmune thyroid cases, have of feeling better on Armour® over Synthroid® may not just be due to the T3 or bioidentical component, but rather to a sensitivity to the fillers in Synthroid® (especially cornstarch). Therefore, when looking at the appropriate thyroid hormone replacement beyond TSH, filler sensitivity may need to be accounted for.

A second issue that is common in practice is the use of bioidentical versus synthetic. This decision is best made by the patient's and healthcare professional's philosophies. I personally would prefer a bioidentical source, especially since we know they are proven to be consistent in their T4 and T3 ratios (despite Synthroid® marketing). However, it is not uncommon for some patients to feel and do better with synthetic hormones. This was common in patients who had autoimmune antibody titers not only against their thyroid tissue structures (TPO Ab, and TBG Ab), but antibody titers to actual T4 and/or T3 hormones. Their immune systems were not only attacking their thyroid tissue, but also thyroid hormones. These patients did worse with bioidentical sources because it was the source of their immune attack, and better with synthetics. Unfortunately, the actual T4 and T3 antibodies are no longer currently available by commercial laboratories. Despite the objective validation of this mechanism, it should be suspected in patients with autoimmune thyroid disorders, especially Graves' disease.

A third issue that is common in practice is the use of T3 hormones. T3 is the active thyroid hormone and has the greatest impact on metabolism; however, it also has a greater risk for overdosing than T4. Patients who have difficulty converting T4 to T3, called underconverters, typically do better with T3-based replacement. Patients who have certain mechanisms for thyroid receptor-site resistance may also do better with T3-based replacement. However, there are some people who feel easily overdosed whenever they take a T3-based hormone. This is most common in hypothyroid Hashimoto's patients who have constant relapsing immune attacks against their thyroid glands, but not enough to become hyperthyroid. Instead, they release more than usual amount of thyroid hormones into their bloodstream from the tissue breakdown, which causes T3 replacement at times to put them over the edge. These patients typically do better with T4-based drugs.

In summary, the most appropriate thyroid hormone will probably best be determined by an initial thorough work-up followed by some trial and error of different types of replacements. The patient's subjective response should be accounted for as well as routine laboratory testing. The influence of laboratory monitoring should extend beyond an isolated TSH and should also include glucose, lipid markers, CBC with differential, transaminases, creatinine, urinary DPD, etc. It should be strongly pointed out that although thyroid hormone replacement is important in cases of hypothyroidism, it should not be limited to it exclusively. Many factors may influence thyroid metabolism and receptor-site responses beyond taking hormones. These include the influence of cytokines on receptor-site sensitivities, the influence of cytokines on converting enzymes, the influence of cytokines on feedback pathways, the influence of non-thyroid hormones on receptor-site responses, the influence of non-thyroid hormones on converting enzymes, etc. It is the goal of this manual to consider all of these factors when working with a thyroid case.

Assessing Patient Responses to Different Hormones

Many times, thyroid patients will already have a history of attempting different types of thyroid hormones before seeking your evaluation. It is important to ask them if they noticed any significant improvements with any particular type of hormone replacement therapy. Some types of thyroid replacement may have made the patient feel wired, excited, nervous, agitated, etc. Some or all types of replacements may have caused no change at all in their symptoms, but may have normalized their TSH. Some types of thyroid replacement may have had more of a noticeable improvement than others. Some

Please see registered trademark ownership in Appendix 1.

types of replacements may have dramatically improved quality of life. In any case, it is always important to ask patients about their experience with the different types of thyroid hormones they have tried. The goal of this diagnostic exercise is not just to find the most appropriate thyroid hormone, but rather to get a deeper perspective on the mechanism or mechanisms involved with their thyroid. Below are some common clinical scenarios and potential mechanisms for their responses.

1. **No significant response to any thyroid hormones, but normal TSH.** This is a very common response in with patients who have Hashimoto's. The inflammatory cytokine response promotes decreased thyroid proteomic responses, the cytokine elevations disrupt the conversion potentials of T4 to T3, and they also promote global loss of endocrine pituitary feedback loop patterns. No significant response to thyroid hormones may be evident from patients who have extreme cytokine elevations, such as with severe injury, recovery from surgery, cancer, autoimmune diseases, etc. Patients may also have no significant response to thyroid replacement if they are unable to appropriately convert T4 into T3, and they are prescribed a T4-only medication. Lastly, the patient may subjectively feel that she has had no response to thyroid medications because they have multiple overlapping conditions other than thyroid mechanisms.
2. **Only felt better with bioidentical hormones.** This may occur for several reasons. For example, a patient may not be able to convert the T4 into T3 provided with T4-only synthetic medications. He/she may have sensitivities to some of the dyes and fillers that are found in synthetic compounds, but not in bioidentical versions. He/she may actually need T3, instead of isolated T4 thyroid hormones due to receptor-site resistance issues. His/her body's receptor sites in general respond better to bioidentical than synthetic compounds that are similar, but not bioidentical.
3. **Only felt better with T3 combination hormones.** This most likely reflects difficulty converting T4 into T3 efficiently enough for his/her body's needs. It is also common to see this with individuals who have underlying mechanisms for thyroid receptor-site resistance that may be promoted by cortisol elevations, progesterone insufficiencies, elevated cytokines, elevated homocysteine, vitamin A deficiencies, etc. These mechanisms are even more likely if there is consistency in improvement with both isolated synthetic T3 and bioidenticals.
4. **Only felt better with synthetic hormones.** This may occur if the patient has overactive metabolism and cannot tolerate the T3 found in bioidentical thyroid hormones. This occurs at times with Hashimoto's patients who are constantly having relapsing immune attacks against their thyroid, leading to surges of thyroid hormones into their bloodstream from thyroid tissue breakdown. Another example where patients only feel better with synthetic hormones occurs when their immune system is not only attacking their thyroid tissue (TPO Ab and TGB Ab) but their actual thyroid hormones. This occurs many times in Graves' disease and when the thyroid failure is a result of polyendocrine autoimmune disorder.
5. **Felt over-stimulated with T3 or bioidentical hormones.** This may occur if the patient has overactive metabolism and cannot tolerate the T3 found in bioidentical thyroid hormones. This occurs at times with Hashimoto's patients who are constantly having relapsing immune attacks against their thyroid, leading to surges of thyroid hormones into their bloodstream from thyroid tissue breakdown. It may also occur in patients who have elevated amounts of epinephrine in their system from excess nicotine, caffeine, severe psychological stress, constant overtraining, or genetic mechanisms.
6. **Felt fatigued and run down with thyroid hormones.** This is almost always related to sensitivities found in the inactive ingredients of the hormones, such as the dyes and fillers. It may also occur if they have an autoimmune response against their actual thyroid hormones that may occur in Graves' or polyendocrine autoimmune disorders.

Conversion Chart for Thyroid Hormones

Drug	Thyroid Tablets, USP (Armour® Thyroid)	Liotrix Tablets, USP (Thyrolar®)	Liothyronine Tablets, USP (Cytomel®)	Levothyroxine Tablets, USP (Unithroid®, Levoxyl®, Levothroid®, Synthroid®)
Approx. Dose Equivalent	¼ grain (15 mg)	¼		25 µg (.025 mg)
Approx. Dose Equivalent	½ grain (30 mg)	½	12.5 µg	50 µg (.05 mg)
Approx. Dose Equivalent	1 grain (60 mg)	1	25 µg	100 µg (0.1 mg)
Approx. Dose Equivalent	1 ½ grains (90 mg)	1 ½	37.5 µg	150 µg (0.15 mg)
Approx. Dose Equivalent	2 grains (120 mg)	2	50 µg	200 µg (0.2 mg)
Approx. Dose Equivalent	3 grains (180 mg)	3	75 µg	300 µg (0.3 mg)

Please see registered trademark ownership in Appendix 1.

Where a condition is identified or a lab result is outside the normal range: Patient should seek appropriate conventional medical management and drug administration by a physician.

Case Examples

Case 25

A patient was diagnosed with hypothyroidism three years ago and was initially given Synthroid® to manage her condition. She was not noticing improvements and found a physician who placed her on Armour®. When she started taking Armour®, she immediately felt worse. She started to have symptoms of night sweats, inward trembling, and anxiety. What mechanisms must be considered?

Case 26

A patient was diagnosed with hypothyroidism and placed on Levothroid®. She had some improvement, but based on the suggestion of a friend, she switched to Westhroid™ and felt dramatically better. What mechanisms must be considered?

Case 27

A patient has been taking Armour® for more than 20 years. Recently, the formula was changed and due to a shortage, she tried Westhroid™. She felt that Westhroid™ was not having any impact on her. She was able to get back on Armour® three weeks later and then felt worse than ever. She switched to Nature-Throid® and felt better again. What mechanisms must be considered?

Case 28

A patient with a long history of hypothyroidism tells you that she has a very sensitive body and cannot tolerate many foods. She has struggled with finding a thyroid hormone as she was unable to tolerate Synthroid®, Unithroid®, Thyrolar®, and Armour®. She can only tolerate Westhroid™. What mechanisms must be considered?

Case 29

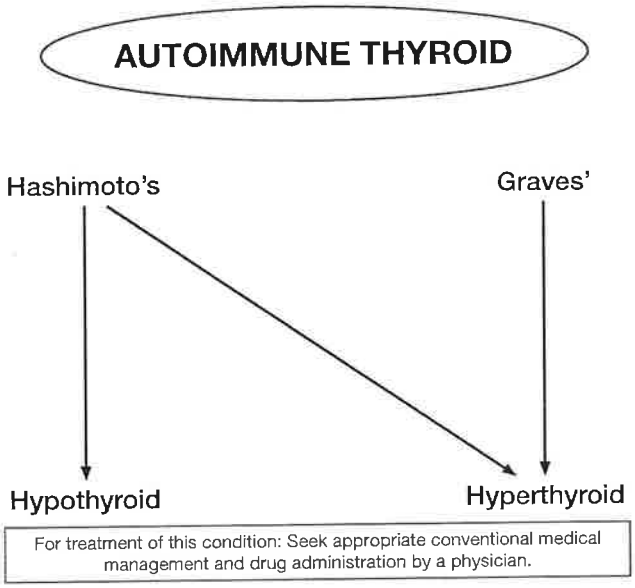
A patient with a history of hypothyroidism tells you that she tried bioidentical hormones like Westhroid™, Nature-Throid®, and Armour® and did not feel well with any of them. She has found that Levoxyl® works best for her. What mechanisms must be considered?

Case 30

A patient tells you that she has had thyroid symptoms the past four years, especially with weight gain. She has been taking Levothroid® during this entire time. She recently changed doctors and was placed on Cytomel® in addition to her Levothroid® and found her metabolism improved, and she was finally able to lose some weight. Yet, she is still struggling with thyroid symptoms. What mechanisms must be considered?

Please see registered trademark ownership in Appendix 1.

Autoimmune Thyroid Connection



Introduction

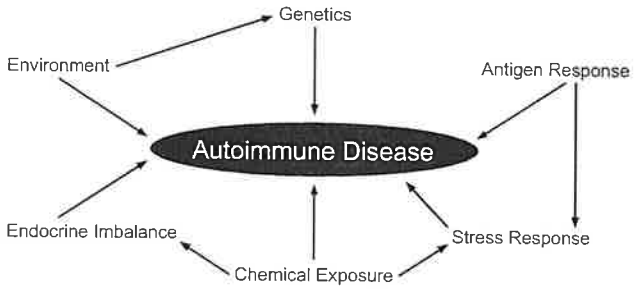
Worldwide, the number one cause of thyroid hypofunction is iodine deficiency, but in the U.S., the number one cause of hypothyroidism is Hashimoto's autoimmune thyroid. Hashimoto's is an autoimmune attack against thyroid cells. It may present as hyperfunction or lead to hypofunction. When the immune attack against the thyroid becomes exaggerated in this condition, it is called Hashimoto's thyroiditis and presents as thyroid hyperfunction (depressed TSH with elevated T3 and/or T4). In these cases, the thyroid's overactive response must be differentiated from Graves' disease. Graves' disease is also an autoimmune condition against the thyroid. The key to distinguishing Hashimoto's thyroiditis from Graves' thyroid hyperfunction is tissue biopsy; however, thyroid antibodies may be useful. In Graves' disease, the thyroid-stimulating hormone (TSH) autoantibodies will be very high, and in an active Hashimoto's thyroiditis, the TPO autoantibodies will be the highest, with or without TSH autoantibodies.

It should be noted, however, that in most cases of Hashimoto's, there is not an active thyroiditis response leading to thyroid hyperfunction. In most cases of Hashimoto's, there is a slow, gradual attack against the thyroid, eventually leading to loss of enough thyroid cells that the condition presents as primary hypothyroidism (elevated TSH). When the thyroid does not work adequately, the pituitary increases the production of TSH as an attempt to increase thyroid gland activity. Most people who have Hashimoto's never actually develop overactive thyroid symptoms. Over time, they start to develop the symptoms of low thyroid function and have their TSH measured. Their TSH will usually be found to be elevated, and they are typically diagnosed as primary hypothyroid and placed on thyroid hormone replacement. The issue of their autoimmune attack is not addressed. Instead, they are considered as being managed by having a normalized TSH from thyroid hormone replacement. In a sense, patients are having their TSH managed, not the underlying mechanism for their thyroid loss. In most of these cases, since their autoimmune response is ignored, over time, they lose more thyroid cells and their requirements for thyroid hormone replacement continue to rise. Even more important, most individuals with Hashimoto's develop hypothyroidism, despite the fact their TSH is managed with replacement, and they continue to have all the symptoms of thyroid hypofunction. Researchers conclude that a "positive correlation between TPO antibody titers and TSH levels in euthyroid subjects suggests that TPO antibodies are indeed a marker of future thyroid failure."[3]

From a functional approach, it is important to identify the mechanism of the autoimmune response and make attempts to modulate its expression. The evaluation of TPO antibodies for case management is not a reliable marker to use for clinical effectiveness since the levels tend to fluctuate during treatment. Furthermore, with some autoimmune responses, if the TH-2 system becomes depressed, B-cell–activity or production becomes compromised. If there is decreased production of B-cell–stimulating cytokines, the TPO antibodies may not be expressed. Clinicians who make statements that they have cured autoimmune thyroid diseases simply by testing thyroid antibodies have a very poor understanding of immunology. It is recommended that all autoimmune thyroid cases be evaluated by a complete immune and cytokine panel to accurately measure the effectiveness of case management. When a regular and periodic evaluation of the immune system is conducted in Hashimoto's, it is quickly evident that most individuals will have long-term and in most cases permanent immune dysregulation. However, effective clinical management of immune expressions can

dramatically hinder the immune response attack on the thyroid in many cases. It should also be noted that many individuals who present with autoimmune thyroid have also been shown to have pernicious anemia and gluten intolerance. A clinical triad has been linked with autoimmune thyroids that includes these other two immune reactions. Assessing serum intrinsic factor antibodies can identify pernicious anemia. Gluten intolerance may be evaluated by gut tissue biopsy, serum transglutaminase, SIgA gliadin, etc.

Mechanism of Autoimmune Disease



For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Hashimoto's Process

Genotypic Susceptibility

↓

Phenotypic Expressions for Hashimoto's

↓

Immune System Slowly Attacks Thyroid Over Several Years

↓

Loss of Thyroid Tissue, Leading to Decreased Thyroid Production and Elevation of TSH

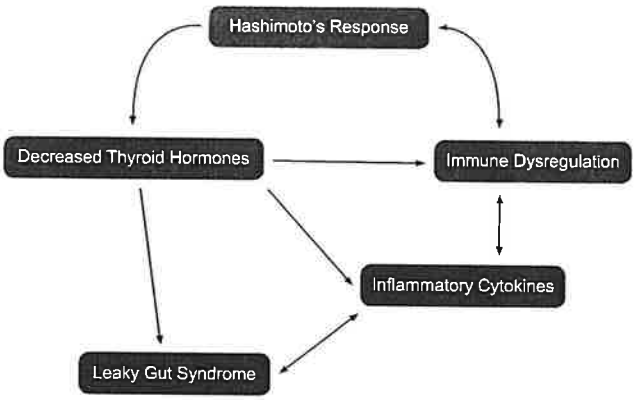
↓

Diagnosis of Hypothyroidism

↓

TSH Managed by Replacement, but No Management for Immune Response, Unless Hyperactive Thyroiditis

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.



For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Thyroid Hormones

↓

Modulate CD4 and CD8 Activity

↓

Modulate TH-1 and TH-2 Expression

↓

Influence the Potential to Regulate Abnormal Autoimmune Expression

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Hashimoto's Autoimmune Thyroid Attack

↓

Destroys Thyroid Gland

↓

Decreases Thyroid Hormone Production

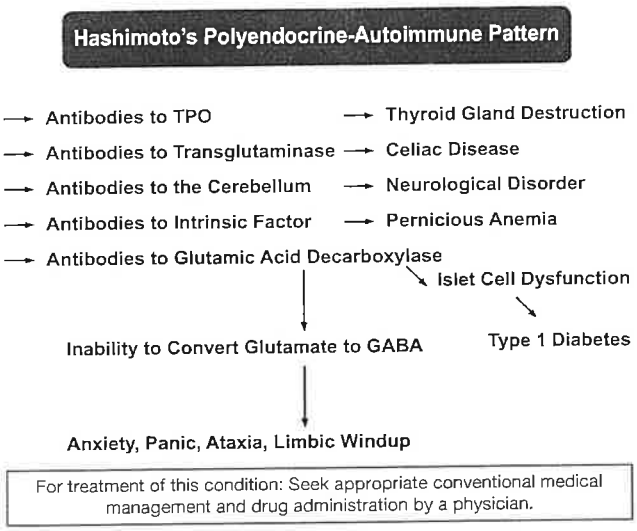
↓

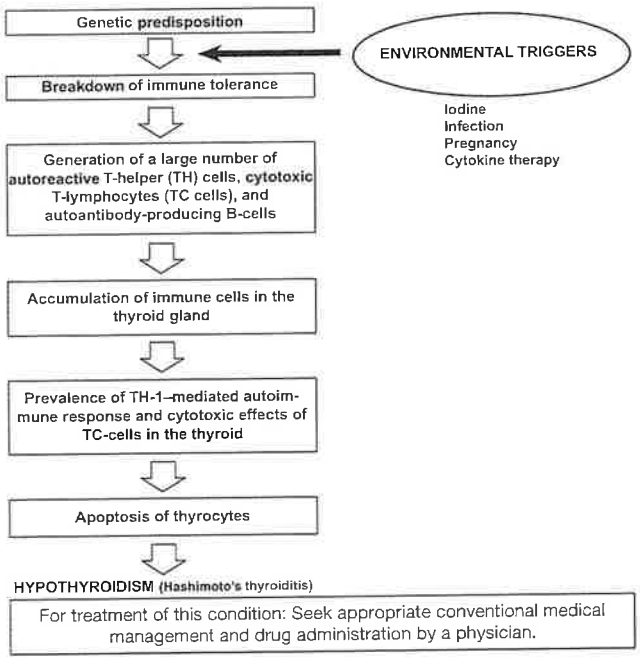
Inability to Modulate TH-1/TH-2 Activity

↓

Promotion and Flare-up of Hashimoto's Response

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.



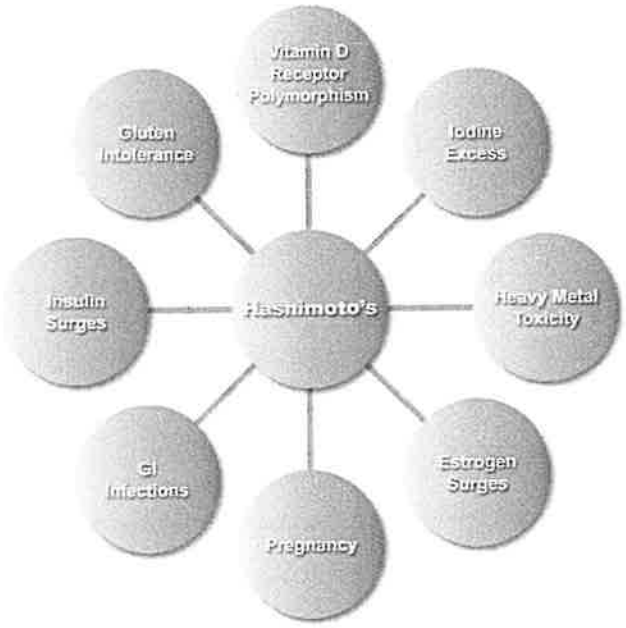


Clinical Considerations with Autoimmune Thyroid

Genetic polymorphisms related to specific antigenic responses, cytokine up-regulation susceptibility, hormone genetic responses, etc, all play a role in the symbiotic interrelationship of physiology that gets expressed as autoimmune thyroid dysfunction.[203]

As with any autoimmune thyroid condition, TH-1 or TH-2 dominance should be established in conjunction with the activity of the immune response. The immune response in relation to patterns of active antigen infection or complete immune dysregulation should be evaluated. In most cases of Hashimoto's thyroiditis, the individual is TH-1 dominant, but this not always the case. There have been several physiological conditions that have been linked to autoimmune thyroid development, which include estrogen dominance, insulin surges associated with PCOS, environmental toxicity, vitamin D deficiency and genetic vitamin D receptor-site polymorphisms, gene susceptibility, stress, and antigen infection. Overall, it appears it is the combination of the above variables that cause certain individuals to turn genes on that start recognizing the thyroid as foreign tissue and creates autoimmune thyroid conditions. Evaluation for metabolic factors associated with the promotion of autoimmune thyroid conditions should be conducted by the clinician. Alteration in NK cell cytotoxicity and production of cytokines might occur in subjects with autoimmune overactivity of thyroid. Studies have suggested that normalization of natural killer cell activity and immune functions could potentially help in the prevention or progression of autoimmunity.

Physiological Factors Promoting Autoimmune Thyroid



For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Insulin Resistance – Polycystic Ovarian Syndrome and Hashimoto's

Polycystic ovarian syndrome has been identified as a potential stimulator of autoimmune thyroid dysfunction.

Polycystic ovarian syndrome is the most common female hormone disorder currently in the United States and is considered the most common cause of infertility. It clinically presents in females as inability to lose weight, hair loss, fatigue after meals, hormone dysregulation, and sugar cravings. Polycystic ovarian syndrome can be identified by patterns of insulin resistance from a standard blood chemistry panel that includes elevated fasting glucose, triglycerides, and cholesterol. The pattern of elevated triglycerides greater than total cholesterol is strongly suggestive of insulin resistance. Insulin resistance promotes the activity of the enzyme 17,20-lyase found in the thecal cells of the ovaries, which leads to increased production of both estrogen and testosterone, but it is the elevated testosterone that suggests the development of PCOS. Diagnostic imaging of the ovaries may reveal multiple ovarian cysts in this condition, but an isolated elevated testosterone with associated symptoms and patterns of insulin resistance is satisfactory in considering PCOS. As insulin resistance stimulates 17,20-lyase, the subsequently elevated testosterone causes the insulin receptor sites to not respond to insulin, further promoting insulin resistance and creating a vicious cycle. These hormone shifts of elevated insulin and testosterone appear to be a promoting factor for the development of autoimmune thyroid dysfunction.[204] Research has demonstrated that insulin surges promote elevated concentration of TH-1 cytokines, which contribute to the promotion of autoimmune thyroid disease.[205] Pregnant women with IDDM who have a positive test for TPO Abs before gestation have poorer glucose control and a high prevalence of autoimmune-triggered hypothyroidism. Therefore, it is recommended that pre-pregnant IDDM patients be screened for anti-TPO antibodies.[206]

Insulin Surge Patterns

Two Mechanisms for Insulin Surges

INSULIN RESISTANCE	HYPOGLYCEMIC
• Feels fatigue after meals • Craves sugar after meals • Has difficulty falling asleep	• Feels energetic after meals • Craves sugar before meals • Has inability to stay asleep • Crashes and craves sweets between 3:00 p.m. and 5:00 p.m.

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Insulin Resistance
For treatment of this condition.
Seek appropriate conventional medical management and drug administration by a physician

Consider diet, nutrition, and lifestyle support for:

- The peripheral utilization of sugars

Hypoglycemia
For treatment of this condition.
Seek appropriate conventional medical management and drug administration by a physician

Consider diet, nutrition, and lifestyle support for:

- Sugar metabolism

Vitamin D and Autoimmune Thyroid

Vitamin D associations have been found in numerous autoimmune diseases, including autoimmune thyroid conditions.[207-208] It appears that vitamin D influences regulatory T-cells, which modulate T-helper expression and the balance of TH-1 or TH-2 dominance.[209-211] It also appears that many individuals who have autoimmune thyroid conditions not only have vitamin D deficiency, but also a genetic polymorphism in their body's vitamin D receptors (VDR polymorphism). To put it simply, they genetically have vitamin D receptor sites that do not respond adequately to normal amounts of vitamin D, and they may require much greater amounts.[212] Vitamin D appears to help against autoimmune-mediated thyroid dysfunction.[207-208]

Many factors promote vitamin D deficiencies, including a lack of sunlight exposure and inadequate consumption of vitamin D–rich foods, such as oily fish. Gastrointestinal inflammatory disorders reduce absorption of vitamin D. Cortisol elevations or use of cortisone can deplete vitamin D levels. Ethnicities with darker skin and obese individuals are more at risk for vitamin D insufficiencies.[213-215] Also, as individuals become older, their efficiency in photoproduction to use sunlight to process vitamin D becomes less efficient.[216-217]

Vitamin D levels may be tested with these individuals with a serum 25(OH)D test, but the ranges used to determine vitamin D status adequacy may not represent individuals with vitamin D receptor-site polymorphisms adequately. Despite measurements of serum vitamin D levels, an autoimmune thyroid pattern may benefit from vitamin D supplementation.

If vitamin D is used, it should be 2,000-5,000 IU per day in an emulsified form.

Altern Med Rev. 2005 Jun; 10(2):94-11
Benefits and requirements of Vitamin D for optimal health: a review.

Vitamin D Serum Levels

Sufficiency Levels
33-80 ng/mL

Insufficiency Levels
20-32 ng/mL

Deficiency Levels
< 20 ng/mL

Gluten Intolerance, Celiac Disease, and Autoimmune Thyroid

Numerous studies have been published linking gluten intolerance to autoimmune thyroid conditions.[3,204,218-220] An evaluation conducted in Europe found "an increased prevalence of celiac disease was demonstrated in adult European and Italian patients with autoimmune thyroid disease [AITD]; conversely, an increased prevalence of autoimmune disease was demonstrated in patients with celiac disease." They concluded their findings by stating that celiac disease is significantly increased in patients with thyroid autoimmune disorders," and that all patients with AITD should be screened for celiac disease.[221]

An investigation between celiac disease and Hashimoto's was conducted in the Dutch population. "A total of 104 consecutive patients with Hashimoto's thyroiditis underwent celiac serological tests (antigliadins, transglutaminase and endomysium antibodies), and HLA-DQ typing. Small intestinal biopsy was performed when any of celiac serological tests were positive. On the other hand, 184 patients with celiac disease were subjected to a thyroid biochemical (thyroid stimulating hormone and free thyroxine), and thyroid serological tests (thyroglobulin and thyroid peroxidase antibodies)." "The data from a Dutch population confirm the association between Hashimoto's thyroiditis and celiac disease. Screening patients with Hashimoto's thyroiditis for celiac disease and vice versa is recommended."[222]

A published case report was of "a 50 year-old woman with autoimmune hypothyroidism of difficult compensation associated with anemia, hypocalcemia with a previous episode of tetany, hypomagnesemia, psychological alterations, and important weight loss. After compensation of the hypothyroidism with doses of L-thyroxine as high as 325 microg/day, the hypothesis of a malabsorptive syndrome was raised. Celiac disease was confirmed by elevated serum antigliadin antibody. A gluten-free diet was instituted which improved the symptoms associated with malabsorption and reduced the L-thyroxine requirement to 125 microg/day." The authors of the case study concluded that "because several studies have shown an association of both diseases, a routine screening for celiac disease has been widely proposed in patients with autoimmune thyroid disease."[223] Another study investigated "the occurrence of clinical and subclinical autoimmune thyroid disease in 79 patients with celiac disease as reflected in thyroid function, antibodies, and ultrasound." They concluded "the occurrence of both clinical and subclinical autoimmune thyroid disease was increased in celiac disease; subclinical thyroid disease indicates regular surveillance."[220]

A study investigated "serum levels of IgA and IgG antigliadin antibodies, IgA antiendomysial antibodies, and IgA anti-tissue transglutaminase antibodies in 169 patients with autoimmune thyroid diseases." They can conclude "that in persons with autoimmune thyropathy there is a high prevalence of positive antigliadin, anti-tissue transglutaminase and antiendomysial antibodies."[224] Other researchers concluded that "minor forms of gluten-induced enteropathy are not rare in patients with thyroid autoimmune diseases, the cost-effectiveness of using antigliadin antibodies as a first-line test (instead of a LT(4) loading test) in patients requiring daily doses of levothyroxine above 2 mcg per kg of body weight, whatever their age."[225]

Molecular mimicry mechanisms with gliadin have been postulated for autoimmune thyroid disorders. The antibodies produced against gluten appear to stimulate B-cell production tagging of the thyroid tissue. Clinically, the associations between gluten intolerance and thyroid dysfunction symptomatology are quite evident. Although different tests are available to evaluate gluten intolerance, the best way to evaluate for these patterns is with the elimination provocation diet. One should be very skeptical of antibody testing for gluten intolerance since many individuals with autoimmune thyroids have compromised TH-2 responding systems. Testing with gut biopsy and serum transglutaminase may offer better options, but elimination/provocation is still considered the gold standard. Many individuals who have

autoimmune thyroid condition present with symptoms of hot flashes, tremors, anxiety, increased heart rate, and increased pulse rate. Many of these symptoms occur when thyroid cells are destroyed and there are increased levels of thyroid hormones in circulation. It is not uncommon to see major resolution of these symptoms in many individuals when they comply with a gluten-free diet.

A study of 100 subjects with autoimmune thyroid disease monitored their response after a gluten-free diet or a gluten-containing diet to evaluate the effect of dietary habits on their thyroid function. The abnormal thyroid serologic markers became undetectable six months after beginning a gluten-free diet. However, thyroid autoantibodies did not positively correlate with dietary habits. This study indicates that a gluten-free diet may have profound impacts on patients with autoimmune thyroid disease, but it does not cure their condition.[226]

Estrogens and Autoimmune Thyroid

Fluctuations of estrogens tend to be a trigger for the gene expression of Hashimoto's in combination with other factors such as cytokine dominance and genetic susceptibility. In addition to potentially turning on Hashimoto's genes, once the gene expression of Hashimoto's takes place, estrogen fluctuations appear to be triggers for the autoimmune response.[227-229] The expression of many autoimmune diseases, including Hashimoto's, develops after pregnancy. This may be due to the extreme shifts of hormones and cytokines that occur during different stages of pregnancy. During the third trimester a pregnant women becomes TH-2 dominant, and during the postpartum state, a TH-1 dominance is promoted. Therefore, clinically many females will state that their thyroid and metabolic disorders developed after they had kids. In these cases, the pregnancy may have promoted their genotype to express the Hashimoto's disease.[229-230]

Another estrogen-susceptible period in a woman's life span besides pregnancy is perimenopause.[231] Many females develop Hashimoto's during perimenopause potentially due to the surges and fluctuations of estrogens. Some of these individuals may develop symptoms from the Hashimoto's response that may crossover clinically to symptoms associated with estrogen fluctuations. During the autoimmune thyroid response, the immune system attacks and destroys thyroid cells that contain thyroid hormones. When these cells are destroyed they release thyroid hormones into the blood stream, which then increase the metabolic rate and produce symptoms such as hot flashes, nervousness, insomnia, and irritability.

Perimenopause and Autoimmunity

Loss of Pituitary-Ovarian Feedback Loop
↓
Surges of Estrogens
↓
Phenotypic Expression of Hashimoto's
↓
Increased Attack of Thyroid Tissue
↓
Abnormal Release of Thyroid Hormones, Increasing Metabolic Rate
↓
Symptoms of Hot Flashes, Insomnia, and Irritability

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Pregnancy and Autoimmunity

Third Trimester → TH-2 Dominance

Postpartum → TH-1 Dominance

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Iodine and Autoimmune Thyroid

Although adequate iodine consumption is important for thyroid hormone production and iodine deficiency is the most common cause of hypothyroidism worldwide, its supplemental use in autoimmune thyroids is contraindicated. Iodine is the major cofactor and stimulator for TPO. Thyroid peroxidase is the enzyme that is under attack with Hashimoto's thyroiditis. It appears that increased iodine intake, especially as a supplement, increases the immune attack on the thyroid.[232] The extreme of this clinical expression is called the Jod-Basedow phenomenon.[233] This phenomenon is portrayed as individuals who are iodine-

deficient in conjunction with elevated thyroid antibodies. When these individuals are given exogenous iodine supplements they develop a hyperfunction autoimmune response. Although this phenomenon does take place in clinical practice at times, iodine supplementation with autoimmune thyroids does not always lead to thyroid hyperfunction. Rather, one observes increased levels of TPO autoantibody levels that multiply dramatically with iodine supplements, and in many instances increased production of thyroid overactive symptoms.

In areas of the world where iodine has been added to sodium chloride (table salt), the rates of autoimmune thyroid have risen. In China, "participants enrolled in a baseline study in 1999, and during the five-year follow-up through 2004, [researchers] examined the effect of regional differences in iodine intake on the incidence of thyroid disease. Of the 3,761 unselected subjects who were enrolled at baseline, 3,018 (80.2 percent) participated in this follow-up study. Levels of thyroid hormones and thyroid autoantibodies in serum, and iodine in urine, were measured, and B-mode ultrasonography of the thyroid was performed at baseline and follow-up." The study concluded that "more than adequate or excessive iodine intake may lead to hypothyroidism and autoimmune thyroiditis."[234]

A study was conducted to evaluate "the evolution of thyroid autoimmunity, in relation to the change in goiter prevalence during 3 years of iodine prophylaxis in Sri Lanka." "These results indicate an evolution of thyroid autoimmune markers during the course of iodine prophylaxis, which has not been described before."[235] A study "evaluated the effects of iodine intake on the prevalence of thyroid dysfunction, autoimmunity, and goiter in two regions with different iodine status after two years of iodization in Turkey. In total, 1,733 adolescent subjects were enrolled into the study." They "measured (free T4), thyrotropin, antithyroid peroxidase antibodies (Anti-TPO), antithyroglobulin antibodies (Anti-Tg), and urinary iodine (UI), and examined the thyroid gland by ultrasound." The research study concluded that "iodine supplementation in Turkey has resulted in the elimination of iodine deficiency in the Eastern Black Sea Region, and this has been accompanied by an increase in the prevalence of autoimmune thyroiditis and thyroid dysfunction."[236]

A study observed "the effects of iodine on the level of CD4/CD8 cells and the production of thyroglobulin autoantibody and thyroid peroxidase autoantibody in Wistar rats," and investigated "the role of iodine in thyroid autoimmunity." The study concluded that "iodine might exert influence on the level of CD4/CD8, and thus the production of thyroid antibodies might directly or indirectly take part in the process of thyroid autoimmunity. Both low iodine and 100 times normal iodine intakes might activate the immune state on some degrees. The effects of iodine on immune responses of TG and TPO antigens in thyroid autoimmunity might not be completely the same."[237] A research study concluded that "thyroglobulin polymorphisms, combined with the explosive mix of iodine, TPO, and H2O2 necessary for thyroid hormone synthesis, inadvertently provide the trigger for the autoimmune thyroid response."[238]

A study evaluated the "prevalence of chronic autoimmune thyroiditis (CAT) and iodine-induced hypothyroidism, hyperthyroidism (overt and subclinical), and goiter in a population exposed to excessive iodine intake for 5 years (table salt iodine concentrations: 40-100 mg/kg salt). This was a population-based, cross-sectional study with 1,085 participants randomly selected from a metropolitan area in São Paulo, Brazil, and conducted during the first semester of 2004. Thyroid ultrasound examination was performed in all participants and samples of urine and blood were collected from each subject. Serum levels of thyroid-stimulating hormone, free thyroxine, and anti-thyroid peroxidase (TPO) antibodies, urinary iodine concentration, thyroid volume, and thyroid echogenicity were evaluated." The study "also analyzed table salt iodine concentrations. At the time the study was conducted, table salt iodine concentrations were within the new official limits (20-60 mg/kg salt). Nevertheless, in 45.6% of the participants, urinary iodine excretion was excessive (above 300 microg/l) and, in 14.1%, it was higher than 400 microg/l. The prevalence of CAT (including atrophic thyroiditis) was 16.9% (183/1,085), women were more affected than men (21.5 vs 9.1% respectively, P=0.02). Hypothyroidism was detected in 8.0% (87/1,085) of the population with CAT. Hyperthyroidism was diagnosed in 3.3% of the individuals (36/1,085) and goiter was identified in 3.1% (34/1,085)." The study concluded that "five years of excessive iodine intake by the Brazilian population may have increased the prevalence of CAT and hypothyroidism in subjects genetically predisposed to thyroid autoimmune diseases. Appropriate screening for early detection of thyroid dysfunction may be considered during excessive nutritional iodine intake."[239]

"It has been reported that hyperthyroidism is associated with an altered endothelial function and increased risk of arterial thromboembolism." A study was conducted "to estimate chosen markers of endothelial dysfunction in iodine-induced thyrotoxicosis (IIT). The groups studied consisted of 41 hyperthyroid subjects, who had been treated with amiodarone (n = 6) or vitamin preparations supplemented with iodine (n = 35) and 40 persons with normal thyroid function. The following parameters were measured: thyroglobulin antibodies (TG Ab), thyroid peroxidase antibodies (TPO Ab), THS receptor antibodies (TR Ab), soluble adhesion molecules: sVCAM-1 and sICAM-1, von Willebrand factor (vWF), plasminogen activator inhibitor-1 (PAI-1), C-reactive protein (CRP), fibrinogen and urine iodine concentration. RESULTS: Patients with IIT had significantly higher levels of sVCAM-1 ($p < 0.01$), IL-6 ($p < 0.005$), fibrinogen ($p < 0.005$) and CRP ($p < 0.05$) in comparison to healthy subjects, whereas sICAM-1, PAI-1 and vWF concentrations did not differ between the groups studied. The highest sVCAM-1 levels were observed in patients with amiodarone induced thyrotoxicosis, and fibrinogen and CRP--in subjects receiving vitamin preparations. There were significant correlations between sVCAM-1 concentration and the levels of sICAM-1 ($r = 0.341$; $p = 0.029$) and PAI-1 ($r = 0.347$; $p = 0.026$), as well as with urine iodine concentration ($r = 0.448$; $p = 0.004$). IL-6 concentration correlated with vWF ($r = 0.456$; $p = 0.003$), TPO Ab ($r = 0.328$; $p = 0.036$) and PAI-1 level ($r = 0.319$; $p = 0.042$)." The study concluded that "iodine induced thyrotoxicosis is associated with an increase of sVCAM-1 and IL-6 levels, possibly reflecting inflammatory and destructive processes in the thyroid gland. However, increased procoagulant activity was not found in patients with IIT."[240]

A study was conducted "to assess the relationship between the biological exposure to iodine and hypothyroidism. METHODS: Logistic regression model was used to analyze the risk factors of hypothyroidism, according to the epidemiologic data of 3,761 adults in 3 kinds of rural communities: mild iodine deficiency area (4 natural villages in Panshan County, Liaoning Province), more than adequate iodine (7 natural villages of Zhangwu County, Liaoning Province), and excessive iodine area (2 natural villages of Huanghua City, Hebei Province)." The study concluded that "more than adequate iodine and excessive iodine were independent risk factors of subclinical hypothyroidism (OR = 3.172 and 6.391, $P < 0.05$) and overt hypothyroidism (OR = 3.696 and 9.213, $P < 0.05$). When interactions of iodine exposure and thyroid peroxidase antibody (TPOAb) or thyroglobulin antibody (TgAb) were included, more than adequate iodine was still a risk factor of subclinical hypothyroidism (OR = 2.788, $P < 0.01$), but had no such effect on overt hypothyroidism. Interaction of more than adequate iodine and positive TgAb significantly affected subclinical hypothyroidism and overt hypothyroidism (OR = 2.656 and 3.347, $P < 0.05$)." In conclusion, "more than adequate and excessive iodine exposure are independent risk factors of hypothyroidism. The risk of hypothyroidism goes up and thyroid dysfunction becomes more serious with the increasing biological exposure to iodine."[241]

"The current iodine status and the impact of silent iodine prophylaxis on the prevalence of autoimmune thyroiditis among schoolchildren in a formerly iodine-deficient community in northwestern Greece were investigated. The findings were compared to those obtained from a similar survey conducted 7 years previously in the same area. A total of 302 schoolchildren (12-18 years of age) from a mountainous area of northwestern Greece were examined for the presence of goiter, and blood and urine samples were collected for assessment of thyroid function, antithyroid antibodies, and urinary iodine excretion. In those children (n = 42) with palpable goiter or positive antibodies and/or a thyrotropin level greater than 5 mU/L, thyroid ultrasonography was performed to estimate thyroid gland size and morphology. Median urinary iodine concentration in the children was 20.21 microg/dL, indicating sufficient iodine intake. Thyroid function was normal in all but 7 children, who had subclinical hypothyroidism (2.5%). Antithyroid antibodies (antithyroid peroxidase [TPO] and/or antithyroglobulin [Tg]) were positive in 32 children, including those with subclinical hypothyroidism (10.6%). Twenty-nine of these children (9.6%) also had the characteristic echo pattern of thyroiditis on ultrasound and were diagnosed to have autoimmune thyroiditis. In comparison to data from our previous survey 7 years ago, there has been a threefold increase in the prevalence of autoimmune thyroiditis among schoolchildren. In conclusion, silent iodine prophylaxis resulted in the elimination of iodine deficiency in Greece, and this has been accompanied by an increase in the prevalence of autoimmune thyroiditis."[242]

"In the mountainous areas of Azerbaijan, the schoolchildren suffer from severe iodine deficiency (ID)

with median urinary iodine excretion (UIE) 36 mcg/l and prevalence of goiter 99% (estimated by US). In a population of 293,000 schoolchildren aged 8-14 y.o.," a study "administered capsules containing 190 mg of iodized oil (Lipiodol-Guerbet, Cedex, France) twice yearly in 6 months apart (total 380 mg). The aim of the present study was to evaluate the efficacy, the benefits, as well as the possible side-effects in a follow-up period of 6 and 12 months after the initial administration of iodized oil. Six and 12 months after the initial administration of iodide, two representative samples of 391 and 326 children respectively were examined. The evaluation included: estimation of goiter by US, determination of UIE and serum measurements of T3, T4, TSH, Tg, autoantibodies against thyroid peroxidase (anti-TPO) and thyroglobulin (anti-Tg)." The results found that "there was an improvement in median UIE, which increased from 36 mcg/l to 68 and 81 mcg/l after 6 and 12 months of treatment respectively. The prevalence of goiter decreased from 99% to 54% and 26% respectively. Tg was decreased at 6 and 12 months from the first administration, whereas TSH remained unchanged at 6 months and decreased at 12 months when compared to the latter value. Hypothyroidism was detected in 7% of children after iodide administration both at 6 and 12 months, but overt hypothyroidism was observed only in 0.5% at 12 months. Subclinical hyperthyroidism was detected in 2% and 6% after iodide administration both at 6 and 12 months. There was a significant increase in the title of thyroid auto antibodies in 6 months, which was retained and increased in 12 months. There was no relation between the appearance of thyroid dysfunction and the positive thyroid auto antibodies." It was concluded that "the dose of 190 mg iodide administered twice yearly, improved iodine deficiency and endemic goiter in schoolchildren. The increase of UIE resulted from iodide administration, was accompanied by an increased title of thyroid auto-antibodies and an increased prevalence of hyper and hypo-thyrotropinemia apparently of no autoimmune etiology."[243]

"Lifelong thyroid hormone replacement is indicated in patients with hypothyroidism as a result of Hashimoto's thyroiditis. However, previous reports have shown that excess iodine induces hypothyroidism in Hashimoto's thyroiditis." A "study investigated the effects of iodine restriction on the thyroid function and the predictable factors for recovery in patients with hypothyroidism due to Hashimoto's thyroiditis. The subject group consisted of 45 patients who had initially been diagnosed with hypothyroidism due to Hashimoto's thyroiditis. The subjects were divided randomly into two groups. One group was an iodine intake restriction group (group 1) (iodine intake: less than 100 micro g/day) and the other group was an iodine intake non-restriction group (group 2). The thyroid-related hormones and the urinary excretion of iodine were measured at the baseline state and after 3 months. After 3 months, a recovery to the euthyroid state was found in 78.3 % of group 1 (18 out of 23 patients), which is higher than the 45.5% from group 2 (10 out of 22 patients). In group 1, mean serum free T4 level (0.80 +/- 0.27 ng/dL at the baseline, 0.98 +/- 0.21 ng/dL after 3 months) and the TSH level (37.95 +/- 81.76 micro IU/mL at the baseline, 25.66 +/- 70.79 micro IU/mL after 3 months) changed significantly during this period ($p < 0.05$). In group 2, the mean serum free T4 level decreased (0.98 +/- 0.17 ng/dL at baseline, 0.92 +/- 0.28 ng/dL after 3 months, $p < 0.05$). In the iodine restriction group, the urinary iodine excretion values were higher in the recovered patients than in non-recovered patients (3.51 +/- 1.62 mg/L vs. 1.21 +/- 0.39 mg/ L, $p=0.006$) and the initial serum TSH values were lower in the recovered patients than in the non-recovered patients (14.28 +/- 12.63 micro IU/mL vs. 123.14 +/- 156.51 micro IU/mL, $p=0.005$). In conclusion, 78.3% of patients with hypothyroidism due to Hashimoto's thyroiditis regained a euthyroid state with iodine restriction alone. Both a low initial serum TSH and a high initial urinary iodine concentration can be predictable factors for a recovery from hypothyroidism due to Hashimoto's thyroiditis after restricting their iodine intake."[244]

Environmental Toxicity and Autoimmune Thyroid

As with any autoimmune disease, the potential role of environmental toxins should always be considered. Antigens that bypass our barrier system and expose themselves to antigen-presenting cells have great potential to create a chronic and powerful immune response. Certain environmental compounds may have greater influences on certain genotypes than others. When evaluating the role of environmental antigens and compounds, referred to as haptens, it is important to consider the immune response rather than just levels. For example, it is likely that most individuals living in society today have mercury compounds in their body. This does not necessarily insinuate that it is a culprit in

the up-regulation of the immune system. What seems to be most important is to identify if the immune system is currently and/or actively recognizing the chemical hapten as foreign. This can be performed by measuring chemical haptens and heavy metal antibodies. For example, if an individual completes a DMSA urine challenge test to screen for heavy metals and the levels of mercury are found to be elevated. That is clinically not as important as the individual with trace amounts of mercury whose mercury IgM antibody levels are elevated. In the second scenario, the immune system has organized a response against the environmental toxin and therefore immune up-regulation is an issue that may be a causative factor in autoimmune physiology. The levels of elevated mercury without antibody elevation are not likely a mechanism of immune up-regulation. In autoimmune disease conditions, especially when the natural killer cell activity is low or the body presents with a pattern of an active antigen by evaluating T-cells, B-cells, and cytokines, a chemical hapten panel and heavy metal antibodies should be considered. If the antibodies are elevated, especially IgM, the immune response related to environmental antigen exposure can be substantiated and attempts to clear the compounds to decrease immune up-regulation should be investigated.

In a recent study, "the impact of dental amalgam removal on the levels of anti-thyroid peroxidase (anti-TPO) and anti-thyroglobulin (anti-Tg) antibodies was studied in patients with autoimmune thyroiditis (AT) with and without mercury antibody responses. Thirty-nine patients with AT were tested by an optimized lymphocyte proliferation test, MELISA, for allergy (hypersensitivity) to inorganic mercury. Patients were divided into two groups: Group I (n = 12) with no hypersensitivity to mercury and Group II (n = 27) with hypersensitivity to mercury. Amalgam fillings were removed from the oral cavities of 15 patients with hypersensitivity to mercury (Group IIA) and left in place in the remaining 12 patients (Group IIB). The laboratory markers of AT, anti-TPO and anti-Tg autoantibodies were determined in all groups at the beginning of the study and six months later." The study concluded that "removal of mercury-containing dental amalgam in patients with mercury hypersensitivity may contribute to successful treatment of autoimmune thyroiditis."[245]

Where a condition is identified or a lab result is outside the normal range: Patient should seek appropriate conventional medical management and drug administration by a physician.

Case Examples

Case 31

A patient with Hashimoto's tells you that when she drinks tea with echinacea or takes immune enhancers, she feels depressed and anxious. It is especially bad when she takes them under stress. What mechanisms must be considered?

Case 32

A patient with Hashimoto's tells you that when she drinks coffee or takes green tea extract, she always feels more energized, and her swelling goes down. What mechanisms must be considered?

Case 33

A patient feels nervous within a few hours of taking the TH-1 stimulator. She also has problems sleeping for the next few days. She tried the TH-2 stimulator and noticed no change in her symptoms. What mechanisms must be considered?

Case 34

A patient notices no change in symptoms when she takes either the TH-1 stimulator or the TH-2 stimulator for three days. What mechanisms must be considered?

Case 35

A patient feels tired, sad, and nervous after taking the TH-1 stimulator or the TH-2 stimulator independently. What mechanisms must be considered?

Introduction to Specific Autoimmune Physiology for Thyroid Dysfunction

Autoimmune responses against thyroid glands are commonplace in clinical practice. Many individuals in the healthcare system present with autoimmune thyroid patterns who are not addressed in case management by healthcare professionals. It is not considered standard practice to investigate autoimmune endocrine dysfunction with individuals who present with glandular defects. The question of why the individual's gland stopped working normally is never questioned.

Clinically, when a patient presents with an autoimmune endocrine pattern, nutritional support specific for the endocrine gland, glandular therapy, and even hormone replacement does very little to improve the individual's response and physiology. In these cases, the clinical focus must be on modulating the immune attack against the gland.

Autoimmune responses can occur against any gland, tissue, hormone, or neurotransmitter. As an introduction to autoimmune endocrine disorders, this section of the notes will elaborate on the most common endocrine gland that is attacked by the body's own immune system, the thyroid gland. Many people are diagnosed with low thyroid activity in the healthcare system, but the mechanism for their glandular default is never conducted. The mechanism for many of these cases is autoimmune. Since the management for these individuals is never addressed, they have very poor clinical outcomes.

When we investigate the mechanisms that lead to an autoimmune response, we see that many variables are involved, which include genetics, environment, endocrine imbalance, chemical exposure, antigen responses, stress responses, etc. The symbiotic orchestration of these factors at a given time appear to switch on genes that present as self-tissue attack by the immune system. One variable in isolation is unlikely to be the sole culprit in the development of autoimmune disease, and therefore an approach that attempts to look for a single variable as treatment will likely prove to be ineffective.

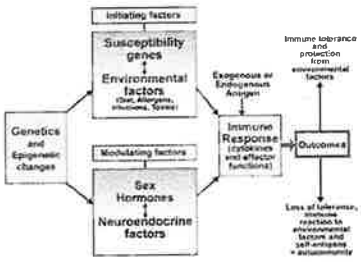
The traditional healthcare system, dominated by a pharmaceutical model, has a limited approach in supporting autoimmune patterns. In traditional models, autoimmune diseases are basically treated with cortisone, other immune suppressing drugs, or thymectomy. The approach in traditional medicine is to suppress the immune system as an attempt to dampen the attack against the self-tissue. Unfortunately, this model creates many side effects and is not clinically effective. Alternative medicine has a varied spectrum of treatment for autoimmune diseases that includes many modalities, but the treatment approach in most of these cases is not centered on specific immune evaluation.

A new model is presented in this section of the manual, which includes specific immune evaluation with natural medicine support. Three clinical attributes are required for this approach: (1) a detailed understanding of immunology is required, (2) an understanding of immunology laboratory testing and interpretation is necessary, (3) an understanding of how natural compounds specifically support the immune system is required. This section of the notes will attempt to provide a general introduction and overview of these attributes beginning with a basic review of the immune system.

Nat Immunol. 2001 Sep;2:(9)777-780.

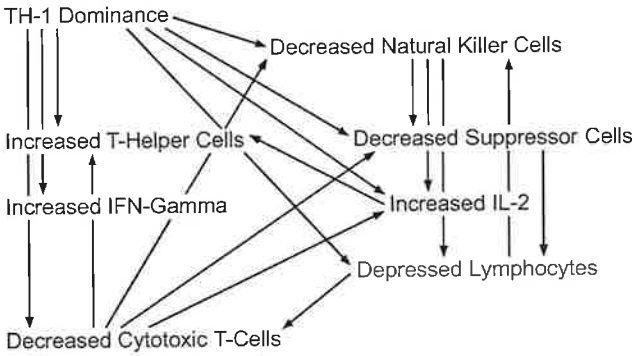
Sex differences in autoimmune disease.

A model for the multifactorial nature of autoimmune diseases.



Sex hormones represent an important modulatory factor in the immune and autoimmune response. Sex hormones include the gonadal sex steroids as well as other hormones that indicate differences between men and women.

Hashimoto's



For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Basic Review of the Immune System

The immune system protects us from invasion by foreign microbes. It is our body's military system to identify and destroy organisms that may harm us or live off our energy reserves, such as bacteria, parasites, viruses, etc. However, the immune system does more than just protect us from the non-self antigens. The immune system is involved with destroying our own cells to help us eliminate dead or senescent cells. This process

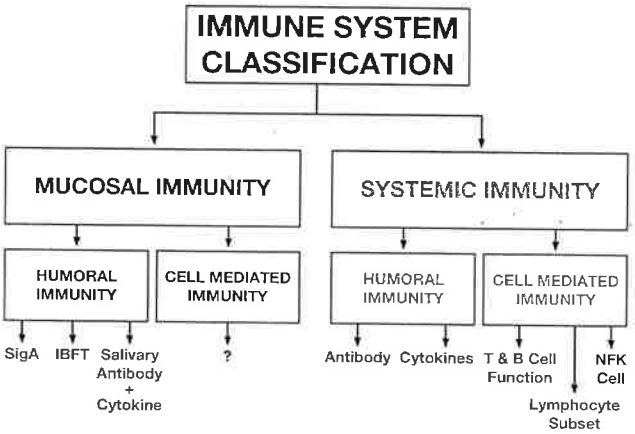
is called cellular apoptosis, and if this system fails, cancer develops. Our immune system is responsible for creating inflammatory responses that allow us to heal after injury, but if the inflammatory response is not regulated properly, it can promote degenerative changes and disease.

Recently, the immune system has been found to interact with the endocrine system. For example, natural killer cells appear to have influence on hypothalamic signaling responses. The immune system response is now recognized as the causative factor in atherosclerosis and cardiovascular disease development. Immunokines are also found to be involved in neurological responses in the central and peripheral neural axis. In addition, neurodegenerative responses are based on immunological responses in specific neuronal tissues.

The immune system's protective function is divided conceptually into what is called specific and nonspecific immunity, which is also known as adaptive and acquired immunity, respectively. Nonspecific immunity involves protection to our body by any organism that is foreign. It is strongly dependent upon macrophage recognition and nonspecific T-cell attack. Specific immunity involves protection against specific antigens. This response requires time for the immune system to develop antibodies to tag the foreign substance to help immune cells identify and destroy the organisms.

The specific immune system is strongly centered upon antibody production by B-cells. Both systems are important for us in order to have ultimate protection. For example, viruses have such a small molecular weight, they cannot often be recognized by macrophages so they can bypass the nonspecific immune system. However, once a virus infects a cell, the immune system can recognize antigenic infiltration and start developing antibodies to help the nonspecific immune system recognize the antigens. The nonspecific immune system is our immediate response against antigenic infiltration of our barrier systems and it is also known as a TH-1 response. In most instances, the TH-1 response is adequate in destroying the organism, before antibodies are produced. If the antigen is not destroyed by the TH-1 system, the specific immune system, also known as TH-2, gets involved.

The TH-2 system produces antibodies to help tag the antigen. Once the antigen has been tagged by an antibody, it is much easier for natural killer cells and cytotoxic T-cells to identify and destroy it. The antibody response takes several days to initiate, and that is why it usually takes several days for the average person to fight the common cold. The TH-1 system is not sufficient in fighting viral infections. The TH-1 system is considered inflammatory and the TH-2 system is considered anti-inflammatory, but these statements are generalizations. In reality, both mechanisms are involved with infectious responses and an absolute spectrum between inflammation and anti-inflammation cannot be established.



Immune Cell Organization

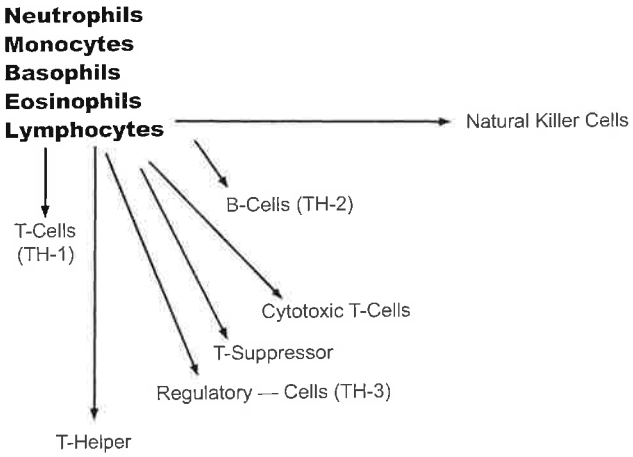
The immune system is comprised of WBCs. These cells are made in the bone marrow, and some mature in the thymus gland. White blood cells are broken into five separate types, which include neutrophils, basophils, monocytes, eosinophils, and lymphocytes. Lymphocytes are further broken down into natural killer cells, B-cells, and T-cells. T-cells are further classified as T-helpers, T-suppressors, regulatory T-cells, and cytotoxic T-cells.

All of these WBCs work together to provide us with protective immunity. Generally, when looking at a complete blood count on a blood chemistry report, the total WBCs will be elevated in acute infections and depressed in chronic infections. Elevations of eosinophils are typically seen with allergies and parasitic infections. Bacterial infections have a tendency to elevate neutrophils, but neutrophil elevations are not specific to bacterial infections. Monocyte elevations can occur with mononucleosis when extremely high, but moderate elevations are not specific to any specific immune response. Monocytes enter tissues and mature into immune cells called macrophages. Macrophages are present in all our tissues and are the immune system's watch guards for antigen infiltration. When an antigen bypasses the body's barrier system (skin, gut

mucosa, lung mucosa), it is typically immediately picked up by a macrophage and immediately attacked. The antigen-macrophage complex becomes known as the "antigen-presenting cell."

Viruses can bypass macrophage antigen surveillance and that is why the TH-2 antibody– producing system is so important for their identification. B-cells are used to make antibodies and are part of the TH-2–specific immune (humoral system). T-cells include T-helper cells, which signal natural killer cells and cytotoxic T-cells to be produced and infiltrate the area of antigen exposure. Natural killer cells and cytotoxic T-cells are the immune system's soldiers involved in attack against the recognized antigen. T-suppressor cells become active when the antigen is destroyed, by sending signals to the other immune cells that the battle is over. Regulatory T-cells are used to balance the activity between T-helper and T-suppressor cells.

White Blood Cells Organizations



The Immune Barrier System

The barrier system of the body is the immune system's first line of defense against foreign microbes. The innate and adaptive immune systems only become involved when the immune barrier system is infiltrated. The barrier system includes the skin, the respiratory mucous membranes, the gastrointestinal mucosa membrane, and the blood-brain barrier. The skin is typically penetrated only by lacerations and trauma; however, the other barrier systems can be compromised by physiological and chemical responses.

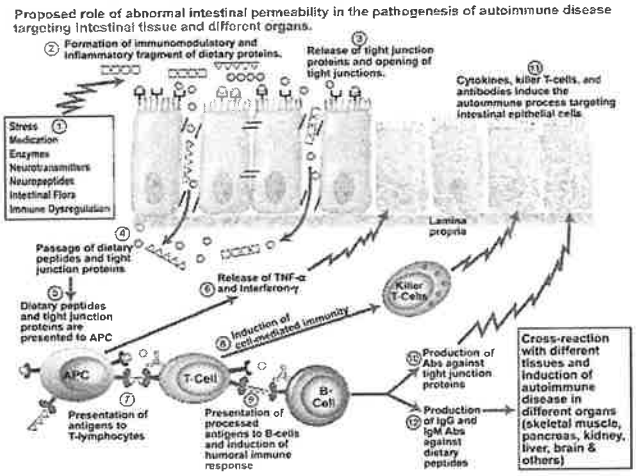
The mucosal barrier systems of the lung and gastrointestinal tract are covered and protected by SIgA cells. These cells serve as a barrier lining for the mucosa, but unfortunately are susceptible to physiological stress responses. The stress response severely hinders the production of SIgA cells and causes damage to the lung and gastrointestinal mucosa barrier system.

The stress response has also been shown to cause loss of blood-brain barrier sensitivity. Clinically, this is important, because if an individual is in a stress response and has antigenic exposure by respiration or by digestion, he or she will be prone to antigen infiltration and immune responses. For example, an individual who works in a mold-infested environment and has suppressed SIgA cells may be at greater risk to respiratory infection than at times when they are not under a stress response. Constant exposure of an antigen to the mucosa can also deplete SIgA cells. For example, chronic parasitic infection may result in low SIgA cells leading to infiltration of other antigens to the gastrointestinal immune system, known as the "gut-associated-lymphoid tissue (GALT)." It is not uncommon to see a history of chronic stress followed by antigenic exposure with individuals suffering from an autoimmune disease.

Physical Barrier System

- Skin
- Respiratory mucous membrane
- Gastrointestinal mucous membrane
- Blood-brain barrier

Intestinal Permeability and Autoimmune Response



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Immune Response Against an Antigen

If an invader is able to penetrate the first line of defense, which is our physical barrier system, then it must face the innate immune system. One of the most important players of the innate immune system is the macrophage. Macrophages are actively seeking out invaders. Foreign invaders (antigens) release chemicals that macrophages are designed to sense. Once these chemicals are perceived by macrophages they seek the foreign invader aggressively and surround the antigen. Once the antigen has been engulfed or surrounded in a pouch, it is called a "phagosome." The macrophage then combines the engulfed antigen vesicle with another vesicle called a "lysosome." The lysosome contains powerful chemicals that are designed to destroy the antigen. This process of destruction is termed phagocytosis.

Macrophages are initially called monocytes when they leave the bone marrow. They are released into the bloodstream, end up at the capillary beds, and then transfer into tissues. This process usually takes about three days. Once in the tissues, the monocytes mature into macrophages. Once the macrophages make it into the tissues, they serve as immune surveillance patrol and stand guard for antigens that have crossed the physical barrier system.

If a macrophage encounters a foreign invader in addition to phagocytosis, it also produces and releases cellular messenger proteins called cytokines. Cytokines call to arms other cells of the immune system to interact in destroying the antigen. In the initial innate immune response, macrophages, complementary cells, and natural killer cells are the main warriors to fight the antigen. The innate immune system will also send cytokines to signal the adaptive immune system to engage. Their response time takes several days because they must produce antibodies for the antigen. Many times, the innate immune system, with the use of macrophages and natural killer cells, will end the fight before the adaptive immune system ever has a chance to attack the foreign invaders. However, with antigens such as viruses, the adaptive immune system becomes very important because the innate immune system is not very effective in dealing with viruses.

The adaptive immune system is responsible for producing antibodies. The adaptive immune system is complex and is capable of producing antibodies against virtually any antigen due to a process called "clonal selection." The function of antibodies is to tag antigens so that the immune system can easily identify them. Antibodies actually do not attack foreign invaders. For example, antibodies can bind to antigens such as bacteria, parasites, and viruses. This process is called "opsonize," which means to "prepare for eating." Once an antigen is bound to an antibody, it signals macrophages to attach to the antibody-antigen complex and perform phagocytosis. The phagocytic process is much more powerful when the macrophage attaches to the antibody-antigen complex then when it attacks the invader alone.

Antibodies perform other vital immune functions other than just tagging antigens and opsonizing them. Antibodies can attach to viruses and prevent them from entering cells. This is important because, once viruses infect cells, they cannot be attacked by the innate immune cells. Antibodies that attach to viruses are called "neutralizing antibodies." The viruses that are no longer able to find safety within cells are then opsonized and ready to be destroyed by phagocytosis.

In order to deal with virus-infected cells that are not threatened by antibody binding, the adaptive immune system sends out killer T-cells. Killer T-cells can actually attack and destroy the virus-infected cells and expose the virus back to the antibodies and the innate immune system.

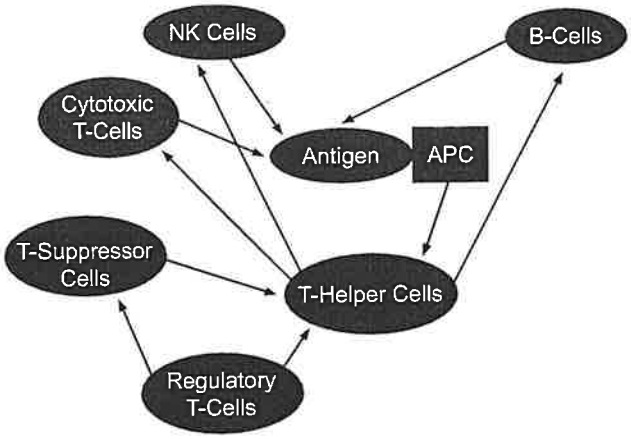
Both B-cells and T-cells are therefore used to deal with antigens such as viruses. It is important to note that there are three types of T-cells. These include killer T-cells, which are also known as cytotoxic T-cells (CTCs), helper T-cells, which are involved with producing cytokines to signal other immune cells, and regulatory T-cells, which help modulate the adaptive immune response.

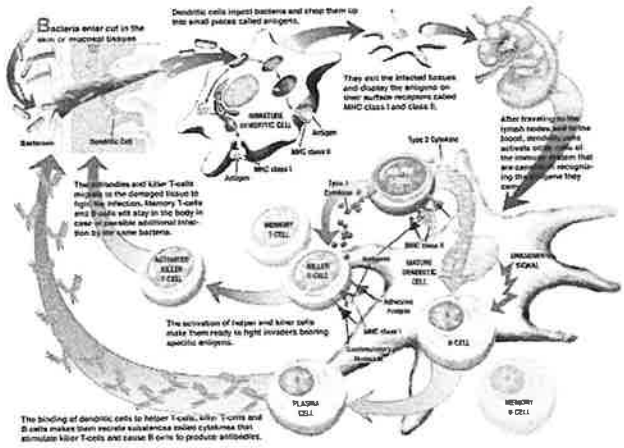
To summarize, antigen exposure to the immune system creates a series of responses. The immune response will be simplified to serve as an introduction to immunology. As the antigen bypasses the barrier system, it comes into contact with macrophages. The antigen is immediately attacked by a macrophage and becomes an "antigen-presenting cell (APC)." The APC releases interleukin-1 which signals T-helper cells to mature and replicate.

T-helper cells release cytokines to activate both TH-1 and TH-2 systems. Interleukin-2 and interferon (IFN) are

used to stimulate the innate immunity. These cytokines increase the production and activity of natural killer cells and cytotoxic T-cells to destroy the antigen. T-helper cells also produce interleukin 4 and 10 to increase B-cell activity and production of antibodies against the antigen. As soon as the antigen is recognized by the macrophage, it is attacked.

Shortly after attack by the APC, the APC sends out signals (IL-1) to T-helper cells, which then send out signals (IL-2 & IFN) to bring in natural killer cells and cytotoxic T-cells to also attack the antigen. As soon as messages are sent to T-helper cells from the APC, signals are also sent (IL-4 & IL-10) to B-cells to produce antibodies. After a few days, if the antigen has not been destroyed by the TH-1 system, antibodies are produced by B-cells and start attaching to the antigen. Once the antigen is tagged with an antibody protein, it is easily recognized and destroyed by natural killer cells and cytotoxic T-cells.





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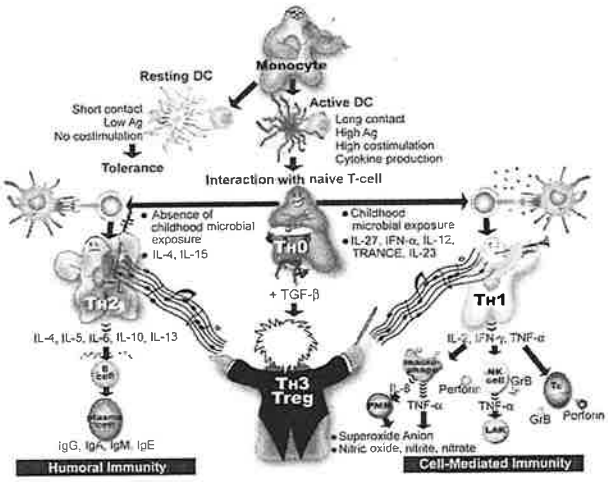

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Regulation of the Immune System by T-Reg Cells

Infectious Agents Associated with Thyroid Disease

"Classical autoimmune thyroid disease (AITD) has been shown to be associated with a variety of infectious agents, while infections of the thyroid gland have been shown to be associated with thyroid autoimmune phenomena. However, the causative role of infectious agents in AITD has not been definitively demonstrated in humans, although AITD can be induced in experimental animals by certain viral infections. Infectious agents may induce thyroid autoimmunity by a variety of diverse mechanisms, such as inducing modifications of self-antigens, mimicking self-molecules, inducing polyclonal T-cell activation (for example by superantigens), altering the idiotypic network, forming immune complexes, and inducing expression of MHC molecules on thyroid epithelial cells. While indirect data suggesting involvement of the infecting organisms in the pathogenesis of human AITD is abundant, only a limited number of studies have employed direct approaches."[246]

These infections are not a conclusive list and in many cases a specific infectious agent cannot be attributed as the sole cause of an autoimmune response.247 Autoimmune thyroid disorders are triggered by a complex set of phenotypic influences with susceptible genotypes. However, the field of immunology has certainly identified specific antigens that promote thyroid dysfunction.248 Additionally, management of the infectious agent does not typically turn off the autoimmune response after it has been triggered, but in certain cases, it may dampen the intensity of the reaction.